TerkHealth360 (Faith Foundation Hospital Management System)

Complete Master Installation Guide & Role-Based Departmental Operational Manual

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Document Information & System Metadata

- **System Version**: TerkHealth360 Enterprise v1.0.0 (Release 2026.1)
- **Deployment Architecture**: Local-First Hybrid EMR / Electron Desktop Client / High-Availability LAN Core
- **Hardware Profile Auto-Adaptation**: Full Support for Dedicated Servers, Apple Silicon MLX, Dedicated NVIDIA/AMD GPUs, and Standard CPU-Only Laptops (e.g., Dell Intel Core i5 with Integrated Graphics)
- **Compliance Standards**: NDPA (Nigeria Data Protection Act), WHO ICD-10 Classification, HL7 FHIR v4 Interoperability, NHIA (National Health Insurance Authority) Act 2022

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1. Master Installation & Deployment Guide (Windows, macOS, Linux)

TerkHealth360 is built as a native multi-platform desktop application using Electron, powered by a high-throughput Node/Express backend and a responsive React frontend.

1.1 Windows Desktop Installation (.exe)

Target Operating Systems: Windows 11, Windows 10 (64-bit editions).

Step 1: Generating the Windows Installer (For Hospital IT Administrator)

On the build machine or central server:

```
cd /path/to/TerkHealth360/frontend
npm run dist:win
```

This produces `release/Faith Foundation Hospital Management System Setup 1.0.0.exe`.

Step 2: Layman Installation on Windows Client Laptops / Desktops

1. **Copy the Installer**: Transfer `Faith Foundation Hospital Management System Setup 1.0.0.exe` via the hospital USB flash drive or download it from the hospital internal network share (`\\192.168.1.100\HospitalApps`).
2. **Double-Click Setup**: Run the `.exe` file.
3. **Handle Windows Defender / SmartScreen**:
4. If Windows displays a blue screen stating: *"Windows protected your PC — Microsoft Defender SmartScreen prevented an unrecognized app from starting"*:
5. Click the small underlined link: **More info**.
6. Click the button: **Run anyway**.
7. **Automated Setup**:
8. The installer automatically creates application binaries in `%LocalAppData%\Programs\Faith Foundation Hospital Management System`.
9. A blue Hospital Cross icon will appear on the **Windows Desktop** and in the **Start Menu**.
10. **Launch**: Double-click the desktop shortcut to launch TerkHealth360.

1.2 Apple macOS Installation (.dmg / .app)

Target Operating Systems: macOS 12 (Monterey), macOS 13 (Ventura), macOS 14 (Sonoma), macOS 15 (Sequoia) on Apple Silicon (M1/M2/M3/M4) and Intel Macs.

Step 1: Generating the macOS Installer (For Hospital IT Administrator)

```
cd /path/to/TerkHealth360/frontend
npm run dist:mac
```

This outputs `release/Faith Foundation Hospital Management System-1.0.0.dmg`.

Step 2: Layman Installation on Apple MacBooks / iMacs

1. **Mount DMG**: Double-click `Faith Foundation Hospital Management System-1.0.0.dmg`. A window opens showing the **Hospital App Icon** and the **Applications** folder shortcut.
2. **Drag and Drop**: Drag the `Faith Foundation Hospital` icon into the **Applications** folder.
3. **Eject Disk Image**: Right-click the mounted disk image on your desktop and select **Eject**.
4. **First Launch (Gatekeeper Verification)**:
5. Open Finder -> **Applications** -> Double-click **Faith Foundation Hospital**.
6. If macOS displays *"Apple cannot check it for malicious software"* or *"Unidentified Developer"*:
    1. Open **System Settings** (or System Preferences).
    2. Click **Privacy & Security**.

3. Scroll down to the **Security** section where you will see: *"Faith Foundation Hospital was blocked from use because it is not from an identified developer"*.
4. Click **Open Anyway** and enter your Mac password.

7. **Microphone Permissions (Mandatory for Voicebox AI Dictation)**:
8. When the app launches, macOS will prompt: *"Faith Foundation Hospital would like to access the microphone"*. Click **OK**.

1.3 Linux Desktop Installation (.AppImage / .deb)

Target Operating Systems: Ubuntu 20.04/22.04/24.04 LTS, Debian 11/12, Linux Mint, Fedora, Red Hat Enterprise Linux.

Step 1: Generating the Linux Package (For Hospital IT Administrator)

```
cd /path/to/TerkHealth360/frontend
npm run dist:linux
```

This generates both: - `release/Faith-Foundation-Hospital-1.0.0.AppImage` (Universal Portable Linux Binary) - `release/faith-foundation-hospital_1.0.0_amd64.deb` (Debian/Ubuntu Package)

Step 2: Layman Installation on Ubuntu / Debian Workstations

Option A: Using the Portable AppImage (No installation required): 1. Copy `Faith-Foundation-Hospital-1.0.0.AppImage` to your Home or Desktop folder. 2. Right-click the file -> **Properties** -> **Permissions** tab -> Check **"Allow executing file as program"**. *(Or via terminal: `chmod +x Faith-Foundation-Hospital-1.0.0.AppImage`)*. 3. Double-click the file to immediately launch the hospital system.

Option B: Installing via Debian Package (`.deb`): 1. Open terminal in the download folder and execute: `bash sudo dpkg -i faith-foundation-hospital_1.0.0_amd64.deb sudo apt-get install -f # Fixes any missing system libraries` 2. The application is now permanently installed in `/opt/Faith Foundation Hospital` and searchable in your Linux Application Launcher under **Medical / Office**.

1.4 Connecting Client Laptops to the Central Hospital LAN Server

In standard hospital operation, one primary computer serves as the **Hospital Database & Voicebox Server** (e.g., in the Server Room or Medical Director's Office at LAN IP `192.168.1.100`), while ward and clinic laptops connect over the secure hospital Wi-Fi/Ethernet.

```
flowchart TD
    subgraph HospitalServer["Hospital Main Server (192.168.1.100)"]
```

```
        DB[(PostgreSQL Database)]
        BE["Backend API (:3000)"]
        VB["Voicebox AI Sidecar (:17493)"]
        DB <--> BE
        BE <--> VB
    end

    subgraph ClinicClients["Hospital Workstations (LAN Clients)"]
        W1["Reception Windows PC"] -->|"http://192.168.1.100:3000"| BE
        W2["OPD Doctor Laptop"] -->|"Microphone Audio / Data"| BE
        W3["Ward Nurse Tablet"] --> BE
        W4["Pharmacy Workstation"] --> BE
        W5["Laboratory Computer"] --> BE
    end
```

Configuring the Client Desktop App to Point to the Server:

1. On any client laptop, navigate to the installed directory or check the `frontend/.env.production` file.
2. Set the `VITE_API_URL` to the server's static hospital IP: `env VITE_API_URL=http://192.168.1.100:3000/api`
3. When staff open the app, it instantly connects to the centralized server, meaning patient records entered at Reception are visible in OPD within milliseconds.

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2. Hospital Cross-Cutting Clinical Architecture & Lifecycle Diagram

The diagram below illustrates how a single patient admission flows across all eight hospital departments without duplicate data entry:

```
sequenceDiagram
    autonumber
    actor Patient
    actor Rec as Receptionist
    actor Nurse as Triage Nurse
    actor Doc as OPD Doctor
    actor Lab as Lab Scientist
    actor Pharm as Pharmacist
    actor Cashier as Billing Cashier

    Patient->>Rec: Arrives at Hospital Front Desk
    Rec->>Rec: Search Master Patient Index (MPI) or Register New
    Rec->>Nurse: Checks-in Encounter -> Queued for Triage
    Nurse->>Nurse: Takes Vitals (BP, Temp, Pulse, SpO2, Weight)
    Nurse->>Doc: Vitals Saved -> Patient Moved to Doctor Queue
```

```
    Doc->>Doc: Voice Dictates SOAP Note via Voicebox
    Doc->>Doc: Auto-derives ICD-10 Diagnosis (e.g., B50.9)
    Doc->>Lab: Requests Lab Order (FBC, Malaria MP)
    Doc->>Pharm: Prescribes Medications (ACT, Paracetamol)
    Lab->>Lab: Samples Accessioned & Results Validated
    Lab-->>Doc: Results Instantly Visible in Patient Chart
    Pharm->>Pharm: Verifies Drug Safety & Allots Stock (FEFO)
    Pharm->>Cashier: Bills Transmitted to Ledger
    Cashier->>Patient: Collects Cash / Validates HMO Pre-Auth & Prints Receipt
    Pharm->>Patient: Dispenses Medications with Spoken Audio Guide
```

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3. Core Security, Universal Navigation & Voicebox AI Studio

3.1 Security & Role-Based Access Control (RBAC)

Every staff member is given an authenticated user profile with cryptographic session tokens. - **Data Confidentiality**: Only licensed clinicians can view subjective consultation notes. - **Audit Logging**: Every button click, patient edit, prescription deletion, or billing cancellation is cryptographically recorded in **Audit Logs** with user ID, IP address, and timestamp. - **Screen Lock**: If leaving your computer unattended, click your Avatar at the top right and click **Lock Screen**.

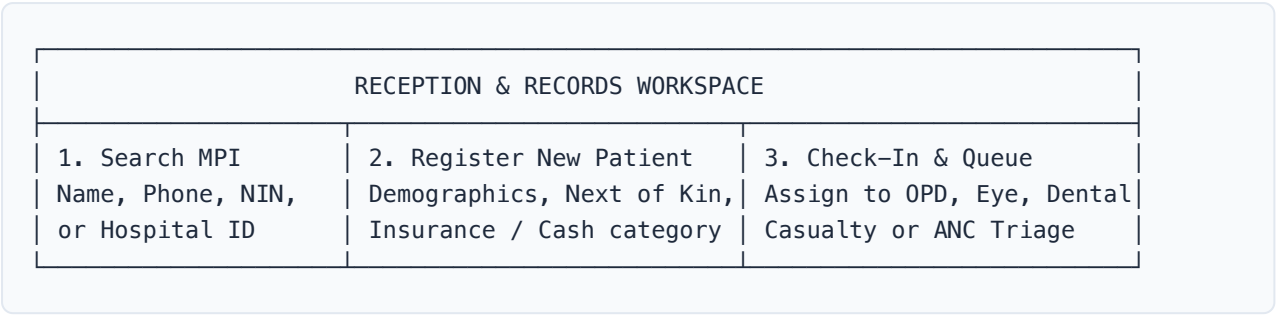
3.2 Voicebox AI Voice Studio (Waveform Icon)

Located in the top header bar next to the emergency banner: - **Zero-Cloud Audio Processing**: All transcription runs inside the hospital premises. No patient voice data leaves the hospital building. - **Hardware-Aware Filter**: Automatically identifies whether the computer has an Apple Silicon chip, an NVIDIA GPU, or is a standard CPU-only laptop (e.g., Dell Core i5). On CPU-only laptops, it restricts models to lightweight **Kokoro 82M** (~350 MB) and **Whisper Small** (~150 MB) to ensure fast operation without overheating.

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4. Section A: Health Records Officers & Receptionists (Front Desk)

Assigned Staff: Medical Records Clerks, Receptionists, Front Desk Executives.
Key Modules: `Patient Registration`, `Patients (MPI)`, `Visits`, `Appointments`, `Queue Management`.



Step-by-Step Operational Instructions:

1. **Searching the Master Patient Index (MPI)**:
2. When a patient approaches the desk, always type their surname, phone number, or national NIN in the top search bar first.
3. *Rule*: Never create a duplicate file if the patient already exists in the system.
4. **Registering a First-Time Patient**:
5. Click **+ New Patient Registration**.
6. Input **Bio-data**: Surname, First Name, Gender, Date of Birth (or estimated age).
7. Input **Contact**: Phone Number, Residential Address, State/LGA of Origin.
8. Input **Next of Kin**: Name, Relationship (e.g., Spouse, Parent), Contact Phone.
9. Select **Payer Category**:
   - *Private Out-of-Pocket Cash*: Standard self-pay.
   - *National Health Insurance (NHIA)*: Enter NHIA Registration Number and select Enrollee Type.
   - *Corporate HMO*: Select HMO Company (e.g., Hygeia, Reliance, AXA Mansard) and enter Policy ID.
10. Click **Save & Generate Hospital Number**. The system issues an automated identifier (e.g., `PAT-2026-00412`).
11. **Queue Check-In**:
12. Click **Start Encounter**. Select the requested clinic: `General OPD`, `Eye Clinic`, `Dental Clinic`, `ANC Maternity`, or `Emergency`.
13. Click **Send to Nursing Triage**. The patient is now queued on the nurse's station screen.

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5. Section B: Nursing Staff (Triage, Inpatient Wards, Labour Ward, Emergency & ICU)

Assigned Staff: Registered Nurses (RN), Midwives (RM), Matrons, Triage Nurses.
Key Modules: `Nursing Dashboard`, `Triage Desk`, `Inpatient Care (IPD)`, `Labour Ward`, `ICU`, `Emergency`.

5.1 Outpatient Triage & Vital Signs Desk

1. Open **Nursing Dashboard** -> **Triage & Vitals Desk**.
2. Select the patient from the **Waiting for Vitals** list.
3. Enter vital sign measurements:
4. **Blood Pressure**: Systolic and Diastolic in mmHg.
5. **Temperature**: In Celsius (e.g., `37.4`).
6. **Pulse Rate**: Beats per minute.
7. **Respiratory Rate**: Breaths per minute.
8. **Oxygen Saturation (SpO_2)**: Percentage (e.g., `98%`).
9. **Weight & Height**: System automatically calculates BMI (kg/m^2).
10. **Automated NEWS2 Clinical Early Warning Scoring**:
11. The system automatically tallies the National Early Warning Score.
12. If SpO_2 drops below 92% or systolic BP exceeds 180 mmHg, the card turns **RED** and sounds a triage alert.
13. Record **Presenting Complaints**: Type or use 🎤 **Voice Dictation** to speak the patient's complaints.
14. Click **Save Vitals & Send to Doctor's Waiting Queue**.

5.2 Inpatient Bed Management & Electronic MAR (e-MAR)

1. Navigate to **Inpatient Care (IPD)** -> **Ward Beds Map**.
2. **Bed Allocation**: Select the ward (*Male Medical*, *Female Surgical*, *Pediatric*, *Maternity*). Click any green bed, search the admitted patient's name, and click **Confirm Bed Allocation**.
3. **Medication Administration (e-MAR)**:
4. Click the patient's active bed card.
5. View scheduled doses ordered by the attending physician (e.g., *IV Ceftriaxone 1g at 14:00*).
6. When administered, click **Administer & Sign**. The nurse's credential and timestamp are logged permanently.

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6. Section C: Medical Doctors & Clinical Specialists

Assigned Staff: Medical Officers, Resident Doctors, Consultants, Ophthalmologists, Dental Surgeons.
Key Modules: `Outpatient Care (OPD)`, `Eye Clinic`, `Dental Clinic`, `Surgical Theatre`, `Specialty Disease Clinics`.

6.1 Clinical Consultation Drawer & Voicebox Dictation (SOAP Workflow)

In **Outpatient Care (OPD)**, select a patient from the **Waiting for Consultation** queue:

```
┌──────────────────────────────────────────────────────────────────────┐
│                        DOCTOR'S SOAP NOTE DRAWER                     │
├──────────────────────────────────────────────────────────────────────┤
│ [🎤 Voice Dictation]   [✨ Smart Auto-Online Assessment & Plan]       │
│                                                                      │
│ [S] Subjective: History of Presenting Complaint (Spoken or Typed)    │
│ [O] Objective:  Vitals review, Systemic & Physical Examination       │
│ [A] Assessment: ICD-10 Diagnostic Codes (Auto-Derived from Symptoms) │
│ [P] Plan:       Clinical Management Guidelines & Patient Instructions│
│                                                                      │
│ [Tabs]: [Prescriptions & Stock]  [Lab Orders]  [Radiology Scans]  [Admit] │
└──────────────────────────────────────────────────────────────────────┘
```

1. **Subjective (S) with Voicebox AI**:
2. Click the blue 🎤 **`Voice Dictation`** button.
3. Speak clearly into your computer microphone: > *"Patient presents with complaints of persistent fever and chills for three days, severe frontal headache, and joint pains. No vomiting or diarrhea."*
4. Click **`Stop Voice Dictation`**.
5. Voicebox Whisper transcribes the audio, automatically strips conversational fillers (`um`, `uh`, stutters), and formats it into formal clinical language.
6. **Objective (O)**:
7. Add targeted examination notes (e.g., *"Chest clear bilaterally, abdomen soft, non-tender"*).
8. **Assessment (A) & ICD-10 Coding**:
9. Click ✨ **Smart Auto-Online Assessment & Plan**.
10. The clinical NLP engine matches symptoms to WHO ICD-10 codes (e.g., `B50.9 — Plasmodium falciparum malaria & febrile illness`, `R51.9 — Headache`).
11. **Plan (P), Prescriptions & Orders**:
12. In the **Prescriptions & Stock** tab: Search drug formulary, select dosage (*1 cap*), frequency (*TDS*), and duration (*5 days*). Click **Add Rx**.
13. In the **Lab Orders & Availability** tab: Check necessary investigations (*Malaria MP*, *Full Blood Count*).
14. Click **Save & Finalize OPD Consultation Note**.

6.2 Eye Clinic (Ophthalmology)

Navigate to **Eye Clinic** (EyeClinic.tsx): - Record **Visual Acuity**: Right Eye (OD) and Left Eye (OS) for uncorrected, pinhole, and best corrected vision. - Record **Tonometry (IOP)**: Intraocular pressure in mmHg. - Document Slit-Lamp Biomicroscopy and Fundoscopy findings.

6.3 Dental Clinic & Odontogram

Navigate to **Dental Clinic** (DentalClinic.tsx): - Interactive **FDI Tooth Chart**: Click on specific tooth numbers (e.g., `#16`, `#46`) to record caries, root canal treatments, extractions, or periodontal pocket depths.

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7. Section D: Pharmacy & Inventory Supply Chain Management (SCM)

Assigned Staff: Pharmacists, Pharmacy Technicians, Store Keepers.
Key Modules: `Pharmacy Dispensary`, `Inventory SCM`, `Drug Safety & Allergies`, `Procurement`.

```
flowchart LR
    Rx["Doctor Prescribes Rx in OPD/Ward"] --> Queue["Pharmacy Dispensary Queue"]
    Queue --> Safety["Allergy & Drug Interaction Check"]
    Safety --> FEFO["Batch & Expiry Allocation (FEFO)"]
    FEFO --> Dispense["Confirm Dispense & Deduct Store Balance"]
    Dispense --> Audio["Generate Spoken Patient Dosage Guide"]
```

Step-by-Step Operational Instructions:

1. **Prescription Verification**:
2. Open **Pharmacy** -> **Dispensary Queue**.
3. Select the patient. The system flags known drug allergies in **RED**.
4. **Batch Allocation & Stock Deduction**:
5. The system uses **FEFO (First-Expiry, First-Out)** principles.
6. Select the active batch from the dropdown. View remaining stock balance and batch expiration date.
7. **Dispensing**:
8. Click **Confirm Dispense & Deduct Stock**.
9. The quantity is deducted from the pharmacy sub-store inventory.
10. **Voice Care Guide**:
11. Click **Generate Spoken Care Guide** to allow Voicebox to generate spoken instructions for the patient regarding drug timing, dosage, and side effects.

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8. Section E: Laboratory Information Management System (LIMS / Pathology)

Assigned Staff: Medical Laboratory Scientists (MLS), Technicians, Pathologists.
Key Modules: `LIMS & Laboratory`, `Pathology`, `Blood Bank`.

1. **Phlebotomy & Sample Accessioning**:
2. Open **LIMS** -> **Sample Collection**.
3. Locate the ordered test. Click **Collect Sample & Generate Barcode**.
4. Affix the printed specimen barcode to the blood/urine container.
5. **Entering Test Results**:
6. Open **Worklist & Processing**. Click on the accessioned test.
7. Enter numeric parameter values (e.g., *WBC: 7.2 x 10^9/L*, *Hemoglobin: 14.1 g/dL*).
8. Values exceeding physiological reference intervals are highlighted with **[H] High** or **[L] Low**.
9. **Validation & Sign-Off**:
10. Click **Validate & Electronically Sign Result**.
11. The result immediately synchronizes with the ordering doctor's OPD and ward chart.

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9. Section F: Radiology & PACS Digital Medical Imaging

Assigned Staff: Consultant Radiologists, Sonographers, Radiographers.
Key Modules: `Radiology & PACS Scans`.

1. **Worklist Execution**:
2. Open **Radiology** -> **Imaging Worklist**.
3. Verify clinical indication and modality (*X-Ray*, *Ultrasound*, *CT*, *MRI*).
4. **Uploading Scans & DICOM Viewer**:
5. Upload DICOM scan files.
6. Use the built-in web PACS viewer to adjust window leveling, zoom, measure distances, and compare anatomical structures.
7. **Drafting Reports with AI Voice Dictation**:
8. Under **Radiographic Findings**, click 🎤 **`Voice Dictation`** to speak your observations.
9. Under **Radiological Impression**, dictate your clinical conclusion.
10. Click **Sign & Release Radiological Report**.

10. Section G: Billing, Cashier, Finance & Health Insurance (NHIA / HMO)

Assigned Staff: Cashiers, Medical Billing Officers, HMO Desk Officers, Hospital Accountants.
Key Modules: `Billing & Cashier Desk`, `Insurance Portal (NHIA/HMO)`, `Finance & General Ledger`.

```
┌──────────────────────────────────────────────────────────────────────┐
│                        BILLING & CASHIER DESK                        │
├──────────────────────────────────────────────────────────────────────┤
│ 1. Aggregate Invoices: Automatically groups OPD, Lab, Pharmacy & Bed fees   │
│ 2. Payment Method: Cash / POS Card / Bank Transfer / Mobile Money           │
│ 3. Health Insurance: Auto-splits 90% HMO Coverage vs 10% Patient Copay      │
│ 4. Output: Prints Cryptographically Signed Audit Receipt                    │
└──────────────────────────────────────────────────────────────────────┘
```

1. **Processing Out-of-Pocket Cash Invoices**:
2. Open **Billing** -> **Pending Payments**.
3. Select patient. The invoice aggregates all consultations, lab tests, and dispensed medications.
4. Select payment channel (*Cash*, *POS*, *Direct Transfer*).
5. Click **Complete Payment & Print Official Receipt**.
6. **National Health Insurance (NHIA) & HMO Pre-Authorization**:
7. Open **Insurance Portal**.
8. Input the **HMO Pre-Authorization Code** received from the insurance desk.
9. The system automatically handles co-payment splits (e.g., 90% HMO liability, 10% patient copay) and prepares claims for monthly reconciliation.

11. Section H: Hospital Administration, HR, Medical Director & IT System Engineers

Assigned Staff: Hospital Administrators, Medical Directors, HR Executives, IT System Administrators.
Key Modules: `Security & IAM`, `User Management`, `Workforce HR & Rosters`, `Infrastructure & Security`, `Settings`.

11.1 Managing Staff Accounts & Permissions

1. Navigate to **Security & IAM** -> **User Accounts & RBAC**.
2. Click **+ Add Staff User**.
3. Enter Name, Official Email, Username, Password, and Role (`DOCTOR` , `NURSE` , `PHARMACIST` , `LAB_SCIENTIST` , `RADIOLOGIST` , `BILLING_OFFICER` , `RECEPTIONIST` , `ADMIN`).
4. Click **Save User**.

11.2 Automated Database Backups & Recovery

Located under **System Architecture** -> **Infrastructure & Security**: - **Automated Daily Backups**: Executes every morning at `02:00 AM` , keeping strictly the **10 most recent backups** and automatically removing older snapshots to preserve disk storage. - **Manual Backup**: Click the yellow **Trigger Backup** button anytime before system maintenance. - **Restore from File**: Click **Restore from File**, select a previously saved `.dump` file, and confirm to restore records.

11.3 Hospital Hard Reset (Factory Wipe)

Located under **Settings** -> **Hospital Deployment Reset**:

> [!CAUTION] Typing `HARD-RESET` wipes operational clinical encounters, inpatient visits, and bills for new hospital commissioning, while preserving your Super Admin login, Drug Formulary, and ICD-10 libraries.

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12. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: The microphone dictation button does not record my voice. What should I check?

1. Ensure your physical microphone is connected and not muted.
2. In Windows Settings -> **Privacy & Security** -> **Microphone**, ensure desktop app microphone access is enabled.
3. On macOS, ensure System Settings -> **Privacy & Security** -> **Microphone** has `Faith Foundation Hospital` toggled **ON**.
4. If accessing via a web browser over the LAN, modern browsers block microphone access on unencrypted `http://` . Ensure you are using the installed **Electron Desktop App** or an `https://` local certificate.

Q2: Can the system run when internet cables are completely disconnected?

Yes. TerkHealth360 is engineered as a **100% local-first hospital EMR**. All database queries, vital signs tracking, e-MAR, billing, and offline Whisper/Voicebox voice transcriptions run inside your hospital LAN with zero internet dependencies.

Q3: How do we back up hospital data to an external hard drive?

1. Open **System Architecture** -> **Infrastructure & Security**.
2. Under **Disaster Recovery (DR) Database Snapshots**, locate the latest snapshot and click the **Download** icon.
3. Copy the downloaded `.dump` file to your external encrypted drive.

Faith Foundation Hospital Management System (TerkHealth360) — Built by Terk-Age LTD.